

Rajarata University of Sri Lanka  
Department of Computing  
ICT2403 – Graphics and Image Processing

GROUP 12

# SEGMENTATION

Final document

Group Members:

- ICT/2023/054
- ICT/2023/055
- ICT/2023/056
- ICT/2023/080

# Introduction

Image segmentation is an important stage in digital image processing. Segmentation means dividing an image into meaningful regions or objects based on properties such as pixel intensity, texture or connectivity. The main goal of segmentation is to separate the region of interest from the rest of the image so that it can be analyzed more easily.

In this project, segmentation is used to detect road degradations such as cracks, potholes and other surface damages from road video frames. The purpose of image segmentation in this system is to separate the damaged regions from the normal road surface. By isolating these image areas, the system can clearly identify where the road defects exist.

Segmentation is applied after the frame enhancement stage raw video frames may contain noise, shadows, and uneven lighting, and low contrast. Frame enhancement techniques such as noise reduction and contrast improvement help to make the damaged areas more visible. When segmentation is performed on these enhanced frames, the algorithm can detect the boundaries in regions of road damage more accurately.

Identifying and isolating road degradation regions correctly is very important for the success of the road damage detection system. If the segmentation process is not accurate, the system may either miss some damage areas or detect normal road regions as damage. Therefore, accurate segmentation helps the system to extract only the relevant damage regions while ignoring the background.

The quality of segmentation also affects the damage severity assessment. After segmentation, the detected region can be analyzed to measure its size, shape or area. These measurements help estimate how serious the road damage is. If the segmentation result is accurate, the severity estimation will also be more reliable.

Segmentation also connects the preprocessing stage in the evaluation stage of the system. After the segmentation process, the detected regions are evaluated using performance matrices such as accuracy, precision, and recall, F1 score, and intersection over union(IoU). These matrices help measure how well the segmentation algorithm detects road damage.

Finally, the segmentation stage is integrated into the complete road damage detection pipeline. In the final system, video frames are first enhanced, then segmented to extract damage regions, and finally, the detected damage is highlighted and saved as output. This integrated system allows automatic detection and analysis of road degradation from video data.

# 1. Proposed Frame Segmentation Pipeline

The frame segmentation pipeline describes the sequence of processing steps used to access road degradation regions from video frames. The purpose of this pipeline is to gradually transform the input road frames into a segmented frame where damage regions such as cracks or potholes can be clearly identified. The proposed pipeline follows several image processing stages including pre-processing, segmentation, and region extraction.

## Segmentation pipeline flow

**Enhancement → Gaussian Blur → Otsu Threshold → ROI → Morphology → Contours → Filtering → Output**

### 2.1 Input Frame

- **Purpose of the Stage:**  
The purpose of this stage is to obtain frames from road videos that will be processed for damage detection.
- **Theoretical Background:**  
Video processing systems analyse individual frames extracted from the video stream. Each frame is treated as an independent image for further image processing operations.
- **Algorithm Used:**  
Frame extraction using a video processing library such as OpenCV.
- **Input:**  
Road video file.
- **Output:**  
Individual road frames.
- **Contribution to Next Stage:**  
The extracted frames provide the raw visual data that will be enhanced and analysed in the following stages.

### 2.2 Frame Enhancement

- **Purpose of the Stage:**  
To improve the visual quality of the frame by reducing noise and improving contrast.
- **Theoretical Background:**  
Image enhancement techniques help highlight important structural features of the road surface while reducing unwanted noise or illumination variations.
- **Algorithm Used:**  
histogram equalization
- **Input:**  
Original road frame.
- **Output:**  
Enhanced frame with clearer road surface features.
- **Contribution to Next Stage:**  
Improved frames make segmentation algorithms more effective when detecting damaged regions.

### 2.3 Grayscale Conversion

- **Purpose of the Stage:**  
To simplify image processing by converting the image into a single intensity channel.
- **Theoretical Background:**  
Most segmentation algorithms rely on intensity differences rather than color information. Grayscale images reduce computational complexity while preserving important structural details.
- **Algorithm Used:**  
Grayscale transformation.
- **Mathematical Representation:**  
$$\text{Gray} = 0.299R + 0.587G + 0.114B$$
- **Input:**  
Enhanced RGB image.
- **Output:**  
Grayscale image.
- **Contribution to Next Stage:**  
Provides intensity-based image data required for segmentation algorithms.

### 2.4 Noise Reduction (Gaussian Blur)

- **Purpose of the Stage:**  
To remove small noise particles that may affect segmentation accuracy.
- **Theoretical Background:**  
Noise in images may cause false detections during segmentation. Gaussian smoothing reduces high-frequency noise while preserving important structures.
- **Algorithm Used:**  
Gaussian Blur filter.
- **Mathematical Basis:**  
Gaussian function:  
$$G(x,y) = (1 / 2\pi\sigma^2) e^{-(x^2 + y^2) / 2\sigma^2}$$
- **Input:**  
Grayscale image.
- **Output:**  
Smoothed image.
- **Contribution to Next Stage:**  
Noise reduction improves the quality of the segmentation process.

### 2.5 Threshold-Based Segmentation

- **Purpose of the Stage:**  
To separate potential road damage regions from the normal road surface based on pixel intensity.
- **Theoretical Background:**  
In this implementation, inverted thresholding is used so that darker damaged regions appear as foreground.
- **Mathematical formula:**  
$$g(x,y) = \begin{cases} 1 & \text{if } f(x,y) > T \\ 0 & \text{otherwise} \end{cases}$$

- **Algorithm Used:**  
Otsu Thresholding
- **Input:**  
Smoothed grayscale image.
- **Output:**  
Binary segmented image.
- **Contribution to Next Stage:**  
This stage highlights possible damaged areas which are refined in later steps.

## 2.6 Morphological Operations

- **Purpose of the Stage:**  
To refine the segmented image by removing noise and connecting fragmented regions.
- **Theoretical Background:**  
Morphological image processing improves the structure of segmented objects using operations based on pixel neighbourhood relationships.
- **Algorithms Used:**
  - Closing
  - Opening
- **Input:**  
Binary segmented image.
- **Output:**  
Refined binary mask.
- **Contribution to Next Stage:**  
Produces cleaner regions that can be analysed using connected component analysis.

## 2.7 Connected Component and Contour Detection

- **Purpose of the Stage:**  
To identify and group connected pixels that represent potential road damage.
- **Theoretical Background:**  
Connected components represent groups of pixels that are connected to each other and share similar properties. Contours define the boundary of these connected regions.
- **Algorithm Used:**  
Contour detection using boundary tracing.
- **Input:**  
Refined binary mask.
- **Output:**  
Detected contours representing road damage regions.
- **Contribution to Next Stage:**  
Allows extraction and marking of the damaged areas.

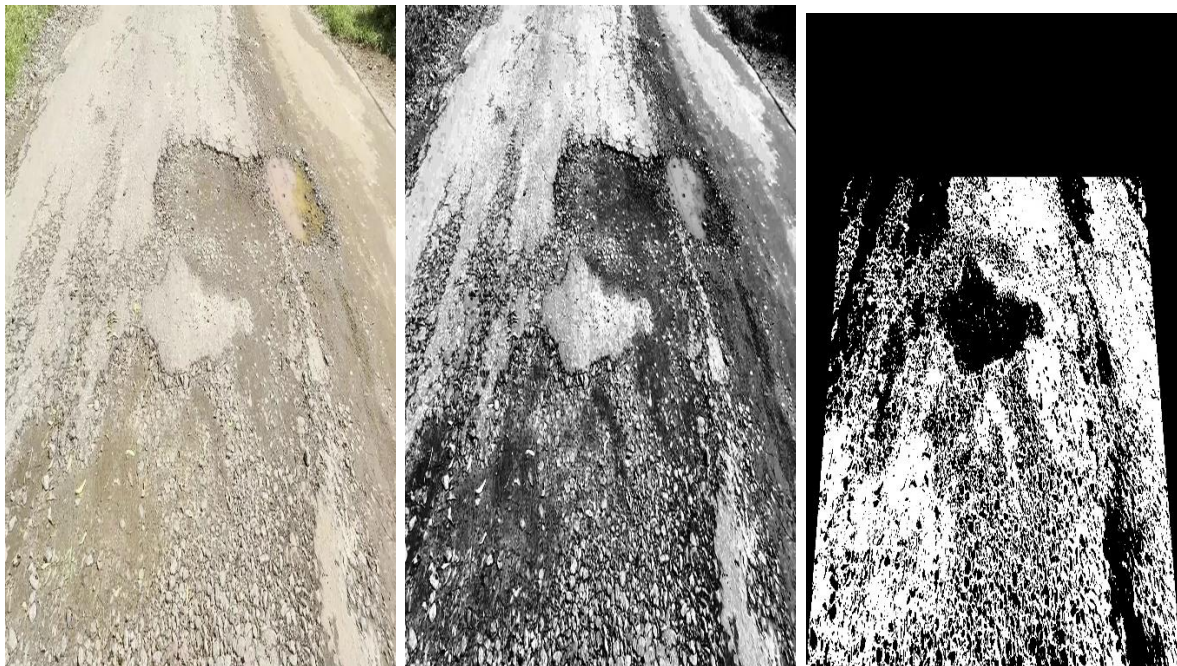
## 2.8 Segmented Road Damage Region

- **Purpose of the Stage:**  
To isolate the regions that represent actual road degradations.
- **Theoretical Background:**  
After contour detection, region filtering techniques such as area-based filtering can remove irrelevant regions.

- **Algorithm Used:**  
In addition, intensity-based filtering was applied to remove bright regions and retain darker damage areas.
- **Input:**  
Detected contours.
- **Output:**  
Final segmented road damage regions.
- **Contribution to Next Stage:**  
These extracted regions are used to mark the detected damage.

## 2.9 Final Marked Output

- **Purpose of the Stage:**  
To visually highlight detected road damage regions.
  - **Theoretical Background:**  
Drawing bounding boxes or contour boundaries helps display the segmentation results on the original frame.
  - **Algorithm Used:**  
Contour drawing or bounding box detection.
  - **Input:**  
Original frame + segmented regions.
  - **Output:**  
Final output frame showing marked road damages.
- Original vs enhanced vs segmented frame comparison





- Contour marking results  
Original vs Contour



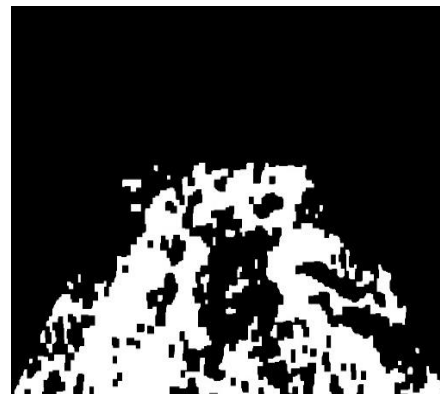
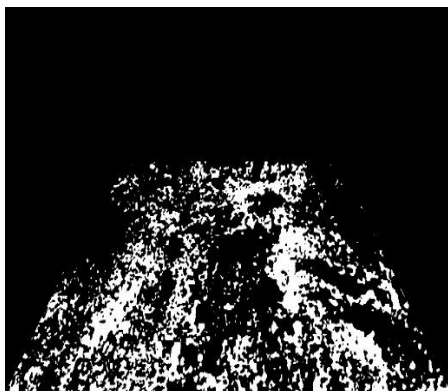
- Morphological refinement examples (if applicable)

Before Morphology vs After Morphology



Threshold Image → Morphology Image

The threshold image contains noise and fragmented regions. After applying morphological operations such as closing and opening, the image becomes cleaner with improved connectivity of regions, which enhances contour detection accuracy.





### 3.1 Test Data Preparation

#### Overview

The test data for this project was prepared from a road video using frame extraction and image processing techniques. The aim was to evaluate how good the segmentation algorithm detects different types of road damage.

#### 1. Number of videos used

- Only one video (Final\_video.mp4) was used in this project.
- The video contains real road conditions with visible damages.

#### 2. Number of extracted frames used for testing

- Frames were extracted automatically using OpenCV.
- Around 699 frames were generated from the video.
- From these, 6 frames were selected for testing and evaluation.

#### 3. Road degradation categories selected

The following types of road damage were considered.

- Potholes
- Cracks
- Raveling

#### 4. Criteria used for frame selection

Frame were selected based on:

- Clear visibility of road damage
- Different lighting condition (bright and shadow areas)
- Different sized of damage (small cracks and large potholes)
- Frames where the road is clearly visible.

#### 5. Annotated ground-truth masks were prepared

- Yes, ground truth masks were prepared manually.
- The predicted mask images were edited using tools like Paint.
- In ground truth:
  - White = damaged area
  - Black = non-damaged area

These were used to calculate evaluation metrics.

#### 6. Frame resolution, format, and preprocessing

- **Resolution:** Same as original video (no resizing applied)
- **Format:**
  - Frames → .jpg
  - Ground truth → .png

#### Preprocessing steps used:

- Conversion to grayscale
- Noise reduction using Gaussian blur
- Segmentation using Otsu thresholding
- Refinement using morphological operations
- Contour detection for final damage extraction

7. **Distribution of frames across different degradation categories**

| Video Name | Frames Extracted | Frames Tested | Road Damage Type      | Lightning Conditions | Remarks                                                                |
|------------|------------------|---------------|-----------------------|----------------------|------------------------------------------------------------------------|
| Video 1    | 483              | 1             | pothole               | Shadow               | Major pothole and cracks detected (Frame 109)                          |
| Video 1    | 483              | 1             | Pothole with raveling | Bright               | Water filled pothole with raveling detected (Frame 421)                |
| Video 1    | 483              | 1             | Cracks                | shadow               | Cracks detected. Some false detection in non-damage areas. (Frame 164) |
| Video 1    | 483              | 1             | Crack                 | Bright               | Most of the Cracks detected (Frame 223)                                |
| Video 1    | 483              | 1             | raveling              | Bright               | Large reveling detected (Frame 366)                                    |
| Video 1    | 483              | 1             | Raveling with pothole | Bright               | Reveling with water filled pothole detected (Frame 416)                |

### 3.2 Evaluation Metrics

The accuracy of the road damage segmentation model was tested through two different evaluation techniques, which included quantitative evaluation and qualitative evaluation. While quantitative evaluation focuses on the numerical precision of the segmentation algorithm, qualitative evaluation considers the visual quality of the segmented road damage regions. The results obtained from the road damage segmentation model were compared against ground truth masks to evaluate the accuracy of the algorithm in detecting road damage.

## Evaluation Metrics Terms

These terms were utilized to compute the evaluation metrics:

True Positive (TP): Road pixels marked as damage that are actually damaged.

True Negative (TN): Road pixels marked as normal that are actually not damaged.

False Positive (FP): Road pixels marked as damaged that are actually normal.

False Negative (FN): Road pixels marked as normal that are actually damaged.

### 3.3 Quantitative Testing on Test Data

Testing was done quantitatively on selected road frames, which had potholes, cracks, and ravelling. The frames were then subjected to segmentation using the full workflow: enhancement, converting the image into grayscale, applying Gaussian blur filter, Otsu's threshold, morphological transformations, and contour filtering. The segmented output was then compared with ground truth masks prepared manually.

The tests were done using Python, OpenCV, and NumPy. The parameters for the test were set as follows:

- Gaussian Blur:  $5 \times 5$  kernel
- Otsu Binary Inverse Thresholding
- Morphological Kernel:  $5 \times 5$
- Morphological closing and opening
- Contour Area Filtering (minimum area)

For evaluating the performance of the algorithm, confusion matrix-based evaluation was conducted to obtain the values of:

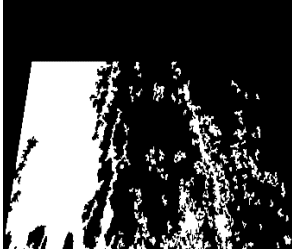
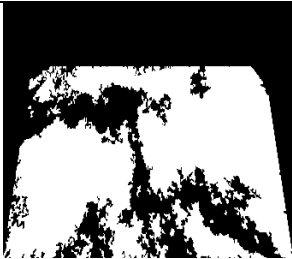
- Accuracy
- Precision
- Recall
- F1-score
- Intersection Over Union (IoU)
- Dice Coefficient

The test was conducted on a frame-wise basis, and the results are shown in Table A.

**Table A – Overall Quantitative Performance**

| Frame | Accuracy | Precision | Recall | F1 Score | IoU    | Dice   |
|-------|----------|-----------|--------|----------|--------|--------|
| 109   | 0.9006   | 0.9674    | 0.8044 | 0.8784   | 0.7831 | 0.8784 |
| 164   | 0.9149   | 0.7927    | 0.9477 | 0.8633   | 0.7595 | 0.8633 |
| 223   | 0.8804   | 0.7279    | 0.8085 | 0.7661   | 0.6208 | 0.7661 |
| 366   | 0.8401   | 0.6979    | 0.9559 | 0.8068   | 0.6761 | 0.8068 |
| 416   | 0.837    | 0.7113    | 0.9759 | 0.8228   | 0.699  | 0.8228 |
| 421   | 0.9323   | 0.9693    | 0.9195 | 0.9438   | 0.8935 | 0.9438 |

### 3.4 Qualitative Results

| frame | original                                                                            | segmented                                                                           | Final output                                                                         | observation |
|-------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|-------------|
| 109   |    |    |    |             |
| 164   |    |    |    |             |
| 223   |   |   |   |             |
| 366   |  |  |  |             |
| 416   |  |  |  |             |
| 421   |  |  |  |             |

## **5. Team Contribution Planning and Individual Responsibilities**

This project was carried out as a team-based assignment, where each member was assigned specific responsibilities related to the segmentation model, experimental evaluation, and final system integration. The objective of this section is to clearly define the individual contributions of each team member and ensure that all tasks were completed collaboratively and efficiently.

At the beginning of the project, the overall work was divided into several key components, including image enhancement, segmentation pipeline development, testing and evaluation, result analysis, and final integration. Each member was assigned a particular road degradation type and was responsible for developing and testing segmentation methods for that category.

### **Task Distribution Strategy**

The work distribution was planned based on the following factors:

- Complexity of different road degradation types
- Individual understanding of image processing concepts
- Coding and implementation skills
- Workload balance among team members

### **Collaboration and Coordination**

The team maintained regular communication to ensure smooth progress. Tasks were discussed and reviewed collectively to maintain consistency in implementation and reporting. Issues encountered during development, such as incorrect segmentation or false detections, were solved through group discussions and testing.

Version control and file sharing were managed using a shared repository, allowing all members to contribute and access the latest updates.

### **Outcome of Team Contribution**

Through effective collaboration and proper task distribution, the team successfully developed a functional road damage detection system. Each member contributed meaningfully to the segmentation pipeline, evaluation process, and final integration. The combined effort resulted in a system capable of detecting multiple types of road degradations from video data.

| Member Name (Reg No)           | Segmentation Responsibility (Degradation Type) | Coding Contribution                                                                                           | Experimental Task                                                                                 | Evidence to Attach                                                          |
|--------------------------------|------------------------------------------------|---------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| D.R.C.L. Bandara (6042)        | Ravelling                                      | Implemented segmentation logic for ravelling detection including thresholding and morphological refinement    | Tested segmentation on ravelling frames under different lighting conditions and evaluated results | Sample frames, segmented outputs, and evaluation results for ravelling      |
| Fathima Rahuma (6043)          | Potholes                                       | Developed segmentation pipeline for pothole detection and contributed to contour filtering logic              | Tested on pothole frames and calculated performance metrics (IoU, F1-score, etc.)                 | Pothole segmentation outputs and performance evaluation screenshots         |
| S.T.A.D.D.N. Wijesinghe (6044) | Alligator Cracking                             | Implemented segmentation for complex crack patterns and contributed to ROI masking and filtering improvements | Conducted testing on crack frames and analysed detection accuracy and errors                      | Ground truth masks, crack segmentation outputs, and comparison results      |
| A.M.M. Amjad (6063)            | Integration and Evaluation                     | Integrated enhancement, segmentation, and output modules into a complete pipeline                             | Performed overall system evaluation and verified performance across all degradation types         | Final output results, integrated system screenshots, and evaluation summary |



## **7. Final Integration Program in Addition to the Segmentation Algorithm**

### **7.1 Integration of the Full Processing Pipeline**

In this final stage of the project, all the components we developed earlier were combined into one complete system. The main goal of this integration is to take a road video as input and automatically detect road damages such as cracks and potholes, then show them clearly in the output.

The system works step by step. First, the input video is loaded and frames are extracted one by one. Each frame is then sent to the image enhancement stage. In this stage, we improve the quality of the image by adjusting contrast and reducing noise. This helps to make the damaged areas more visible, especially in cases where the lighting is not very good.

After enhancement, the frame goes through the segmentation process. Here, the system tries to identify damaged regions on the road. We used basic image processing techniques such as converting the image to grayscale, applying blur to remove noise, and using thresholding to separate damaged areas from the background. Morphological operations were also applied to clean up the segmented result.

Once the segmentation is done, the detected regions are marked on the original frame. We used contour detection to find the boundaries of the damaged areas. Small or irrelevant regions were ignored, and only significant areas were highlighted. This makes it easier to visually understand where the damage is located.

Finally, the system saves the results. For each frame, we store the enhanced image, the segmented mask, and the final output with marked damage. These outputs are useful for both evaluation and reporting.

Overall, integrating all these steps helped us create a simple but functional system that can process road videos and highlight damaged areas.

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### **7.2 Final Deployment Code**

The final system was implemented using Python. We mainly used OpenCV for image processing and NumPy for handling arrays and calculations. The code was organized into separate files to make it easier to manage and understand.

Project Structure

The folder structure of the project is shown below:

```
project/
| — final.py
| — enhanced/
| — threshold/
| — morphology/
| — masks/
| — output/
```

The main.py file is used to run the whole system. It connects all the steps together. Other files handle specific tasks. For example, enhancement.py is used for improving image quality, segmentation.py is used for detecting damaged regions, and visualization.py is used for marking those regions on the image.

#### Libraries Used

The following libraries were used in this project:

- OpenCV
- NumPy
- Matplotlib (only for displaying some results)

#### How to Run the Program

To run the system, the following steps can be followed:

1. Install required libraries:
2. pip install opencv-python numpy matplotlib
3. Place the input video inside the input\_videos folder
4. Run the main file:
5. python main.py
6. The outputs will be saved automatically in the output folders

#### Inputs and Outputs

- Input: Road video or extracted frames
- Output:
  - Enhanced images
  - Segmented masks
  - Final images with marked damaged areas

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### 7.3 GitHub Repository Update

All the project files were uploaded to a GitHub repository so that the work can be easily shared and accessed. The repository includes everything needed to understand and run the system.

It contains the source code, input videos, extracted frames, enhanced images, segmentation results, and final outputs. We also included some sample results to show how the system performs in different conditions.

A README file is also added to explain how to run the project and what each folder contains. This makes it easier for anyone to use or review the project without confusion.

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#### **7.4 GitHub Repository Link**

The GitHub repository link is given below:

*<https://github.com/Fthym-Rhm/road-damage-enhancement.git>*